Root Cause Analysis Report

Part	MS-4505-001 Sheet Metal Bracket
Material / process	SPCC Steel / bending
Case	rca_7b31e3b7dc634aed89b92d14e9bf80dc
Date	2026-11-01 17:54:18
Status	AI proposal, pending expert review

1. Problem

Defect	Severity	Confidence	Detail
scratch	minor	95%	46.1 mm long
dent	major	88%	~14 mm
deformation	major	99%	bend 93.1 deg (nom 90 deg ±0.5)

2. Root cause

Cause: Tool Wear and Die Damage

Reason: The inspection identified minor scratches and major dents on the SPCC steel bracket. According to evidence [seed-003], tool wear and die damage directly cause scratches, dents, and dimensional variations.

Effect: Scratches, Dents, Dimensional Variations

Confidence: 67%

Supporting evidence

- **seed-003**: Tool Wear and Die Damage cause scratches, dents and dimensional variations. Preventive actions: regular tooling inspection, preventive maintenance schedule, die regrinding.

3. Solution

Corrective actions (fix this occurrence)

1. Inspect tooling for damaged surfaces or debris
2. Perform die regrinding or tool replacement

Preventive actions (stop it recurring)

1. Establish a regular tooling inspection schedule
2. Implement a preventive maintenance schedule

Precautions

- Monitor die and tool surface conditions prior to starting production runs

How to verify the fix

Inspect post-maintenance sheet metal parts to confirm absence of scratches and dents

4. Other causes considered

Rank	Cause	Confidence	Reason
2	Improper Bending Force or Over Bending	64%	A major deformation defect was detected during the bending process. Evidence [seed-001] notes that improper bending force or over bending causes deformation and dimensional inaccuracy.

3	Material Inconsistency and Hardness Variation	61%	Major deformation on the sheet metal bracket can also stem from variations in raw material properties, as evidence [seed-002] links material inconsistency and hardness variation to unpredictable deformation.

5. Traceability

AI model: gemini:gemini-3.5-flash. Expert edited the cause: no.
This analysis was drafted with AI assistance from the retrieved knowledge-base evidence. It must be reviewed and confirmed by a qualified quality expert before corrective action is taken.